

# Chapter 5: Determination of Forward and Futures Prices

## 5.1 Investment vs. Consumption Assets

The method used to price a forward contract depends heavily on the type of underlying asset.

- **Investment Asset:** An asset held for investment purposes by at least some traders. The key feature is that these traders are willing to buy the asset now and hold it for the future.
  - **Examples:** Stocks, bonds, gold, and silver.
  - **Pricing:** We can use **arbitrage arguments** to determine their forward prices precisely. This is because we can compare the cost of "buy-and-hold" versus buying a forward contract.
- **Consumption Asset:** An asset held primarily for use or consumption, not for investment.
  - **Examples:** Crude oil, corn, copper, pork bellies.
  - **Pricing:** We *cannot* use pure arbitrage arguments. The benefits of physically holding a consumption asset (e.g., ensuring a factory has enough copper) cannot be perfectly replicated by a financial instrument, leading to a more complex pricing relationship.

## 5.2 Short Selling

Short selling is a crucial mechanism for arbitrage, as it allows a trader to execute a "sell-high, buy-low" strategy by selling an asset they do not own.

### The Process

1. **Initiate:** An investor instructs their broker to "short" an asset. The broker borrows the asset from another client and sells it in the market. The cash proceeds from this sale go into the investor's margin account.
2. **Maintain:** The investor must pay any dividends or interest that the borrowed asset generates to the original owner. They must also maintain a margin account to cover potential losses if the asset price rises.
3. **Close:** The investor buys the asset back from the market and returns it to the lender, closing the position.

### Cash Flow Example

A short sale creates cash flows that are the mirror image of a "buy-and-hold" strategy.

| Action              | Buy and Hold                | Short Sale                   |
|---------------------|-----------------------------|------------------------------|
| April (Price \$120) | - \$60,000 (buy 500 shares) | + \$60,000 (sell 500 shares) |
| May (Dividend \$1)  | + \$500 (receive dividend)  | - \$500 (pay dividend)       |
| July (Price \$100)  | + \$50,000 (sell shares)    | - \$50,000 (buy back shares) |
| Net Profit          | - \$9,500                   | + \$9,500                    |

## 5.3 Assumptions and Notation

To build our pricing models, we start with a set of ideal assumptions for at least some key market participants (like large dealers):

- No transaction costs.
- The same tax rate on all profits.
- Borrowing and lending occur at the same risk-free rate.
- All arbitrage opportunities are taken immediately.

### Notation

- $T$ : Time to maturity of the contract (in years).
- $S_0$ : Spot price of the asset today.
- $F_0$ : Forward price of the contract today.
- $r$ : The risk-free interest rate (continuously compounded) for maturity  $T$ .

## 5.4 Forward Price for an Investment Asset (No Income)

This is the simplest and most fundamental relationship in derivatives pricing. It applies to assets like non-dividend-paying stocks or zero-coupon bonds.

### The No-Arbitrage Principle

The cost of acquiring an asset via a forward contract must equal the cost of acquiring it today and holding it until the delivery date.

- **Cost of "Buy and Hold":** Buy the asset for  $S_0$  today and finance it by borrowing. The cost to repay at time  $T$  is  $S_0e^{rT}$ .
- **Cost of using a Forward:** Enter a forward contract to buy the asset at time  $T$  for price  $F_0$ .

If  $F_0$  and  $S_0e^{rT}$  are not equal, an arbitrage profit is possible.

### The Pricing Formula

For an investment asset with no income, the forward price is:

$$F_0 = S_0e^{rT}$$

### Arbitrage Example

- **Asset Price ( $S_0$ ):** \$40
- **Risk-Free Rate ( $r$ ):** 5%
- **Time to Maturity ( $T$ ):** 3 months (0.25 years)
- **No-Arbitrage Forward Price:**  $F_0 = 40e^{0.05 \times 0.25} = \$40.50$

#### Scenario 1: Forward Price is too HIGH ( $F_0 = \$43$ )

- **Action:** Borrow \$40, buy the asset, and short a forward contract to sell for \$43.
- **Result:** At maturity, sell the asset for \$43, repay the loan of \$40.50, and lock in a **\$2.50 profit**.

#### Scenario 2: Forward Price is too LOW ( $F_0 = \$39$ )

- **Action:** Short the asset, invest the \$40 proceeds, and go long a forward contract to buy for \$39.
- **Result:** At maturity, your investment grows to \$40.50. Buy the asset for \$39 using the forward contract, return it to close the short, and lock in a **\$1.50 profit**.

Because these arbitrage opportunities exist, the market will force the forward price to be exactly **\$40.50**.